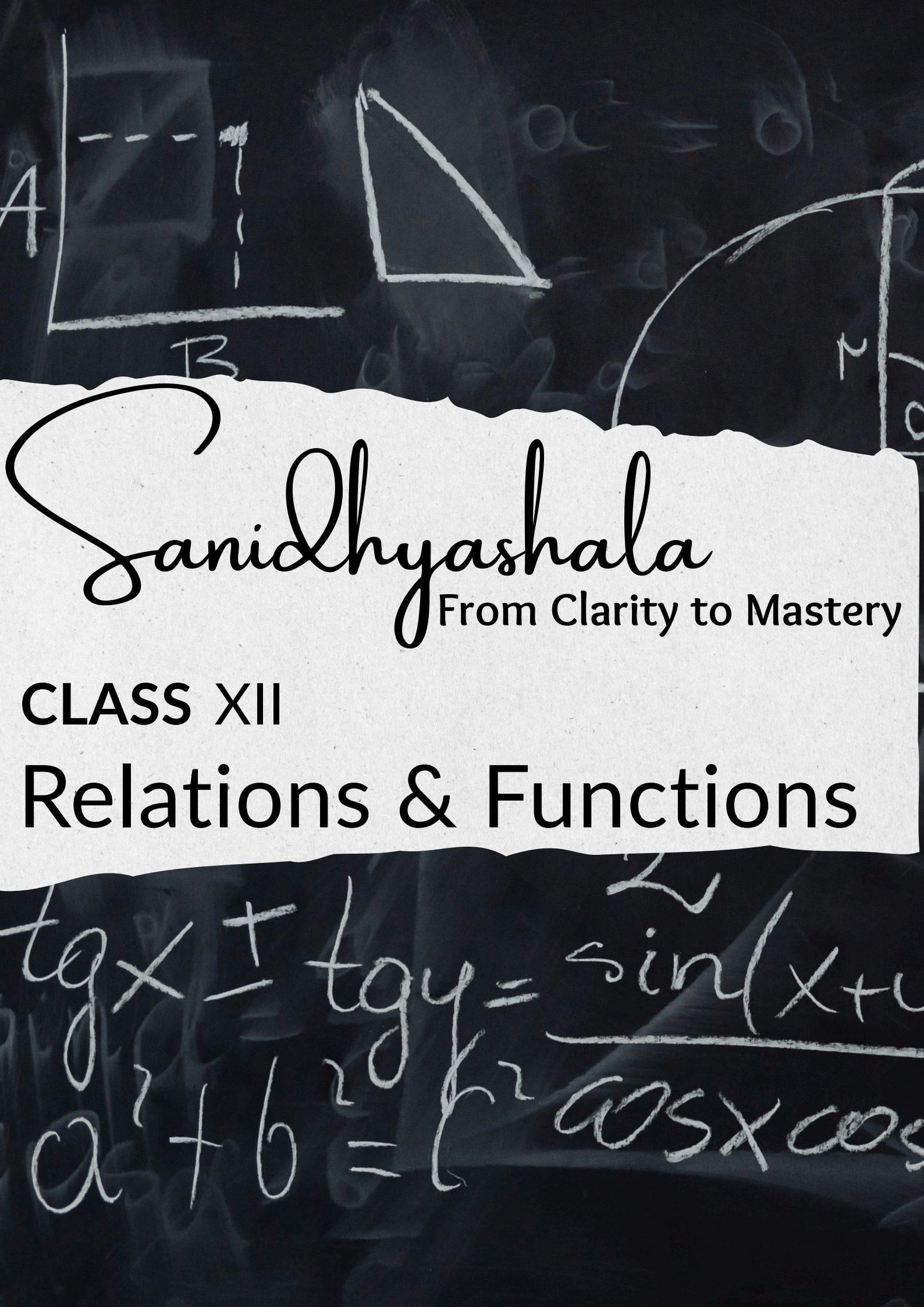




Sanidhyashala

From Clarity to Mastery

CLASS XII

Relations & Functions

$$\tan x \pm \tan y = \frac{\sin(x \pm y)}{\cos x \cos y}$$
$$a^2 + b^2 = c^2$$

NOTES

"Mathematics begins not with answers, but with seeing relationships."

About This Resource

These notes are not designed merely to help you solve examination questions. They are designed to help you understand the ideas that lie beneath those questions.

In many classrooms, Relations and Functions are introduced as a collection of definitions that students are expected to memorise. While definitions are important, mathematics becomes meaningful only when we understand why those definitions exist.

This resource attempts to approach the chapter differently.
Instead of beginning with formulas, we begin with understanding.

Instead of memorising concepts, we explore their meaning.

Instead of treating mathematics as information, we treat it as a language for describing relationships.

NOTES

Why This Chapter Matters

Before mathematics can describe motion, probability, geometry, growth, or change, it must answer a simpler question:

How are two things connected?

A teacher is connected to students.

A city is connected to its population.

A day is connected to its temperature.

A number is connected to another number through a rule.

The language that mathematics uses to describe such connections is called Relations and Functions.

This chapter introduces one of the most fundamental ideas in all of mathematics:

Mathematics is not merely about objects. It is about relationships between objects.

Whenever one quantity depends upon another, functions emerge.

Whenever connections are established between sets of objects, relations emerge.

The ideas learned in this chapter will appear repeatedly throughout calculus, algebra, coordinate geometry, probability, economics, computer science, and data science.

Understanding this chapter well makes many future chapters easier.

NOTES

A Question Before We Begin

Suppose a school maintains a record of students and their roll numbers.

Each student is connected to exactly one roll number.

We may represent this relationship as:

Student $\rightarrow$ Roll Number

Now suppose we create another list:

Student $\rightarrow$ Favourite Colour

Many students may choose the same colour.
Is this relationship different from the first one?

Can mathematics describe both situations?

The answer is yes.

The tools developed in this chapter allow us to study such relationships with precision. This is where our journey begins.

NOTES

Core Concepts

1. Ordered Pairs

Before studying relations, we must understand one of the simplest objects in this chapter: the ordered pair.

An ordered pair is written as:

$$(a, b)$$

The word "ordered" is extremely important.

In an ordered pair, position matters.

For example:

$$(2, 3) \neq (3, 2)$$

Although both pairs contain the same numbers, they represent different arrangements because the first and second positions have different meanings.

Think About It

Suppose a school record contains:

(Student Name, Roll Number)

If we write:

(Roll Number, Student Name)

the information changes.

The same elements are present, but their positions carry different meanings.

This is why order matters.

Equality of Ordered Pairs

Two ordered pairs are equal if and only if their corresponding elements are equal.

$$(a, b) = (c, d)$$

if and only if $a = c$ and $b = d$

Common Mistake

Students often assume:

$$(2, 3) = (3, 2)$$

This is incorrect.

Remember:

The order of elements is part of the information.

NOTES

2. Cartesian Product

Once we understand ordered pairs, we can combine entire sets using them.

Suppose: $A = \{1, 2\}$ & $B = \{a, b\}$

The Cartesian Product of A and B is written as: $A \times B$
and consists of all possible ordered pairs whose first element belongs to A
and second element belongs to B.

Therefore: $A \times B = \{(1,a), (1,b), (2,a), (2,b)\}$

Understanding the Idea

Imagine that every element of A shakes hands with every element of B.

The Cartesian Product records every possible handshake.

That is why every possible pairing appears.

Important Result

If: $n(A) = m$ and $n(B) = n$ then: $n(A \times B) = mn$

In words:

Number of elements in Cartesian Product

=

(Product of the number of elements in each set)

Observation

In general: $A \times B \neq B \times A$

Why?

Because order matters. For example:

$(1,a)$ and $(a,1)$ are different ordered pairs.

Therefore: $A \times B$ and $B \times A$ are generally different sets.

Reflection

Ordered pairs teach us that position carries meaning.

Cartesian Products teach us that mathematics can systematically describe
every possible connection between two collections of objects.

Together, they prepare us for the next and most important idea of this
chapter: Relations.

NOTES

3. Relations – The Language of Connection

Now that we understand ordered pairs and Cartesian Products, we are ready to study the central idea of this chapter. A relation is simply a collection of connections. Whenever we decide which elements are connected and which are not, we create a relation.

Mathematically, if A and B are two sets, then any subset of $A \times B$ is called a relation from A to B . In symbols:

$$R \subseteq A \times B$$

This definition may appear technical at first, but the idea behind it is simple.

A Cartesian Product contains all possible connections.

A relation selects only those connections that satisfy a certain rule.

Understanding Through an Example

Suppose: $A = \{1,2,3\}$ & $B = \{2,4,6\}$ Let us define a relation: " $y = 2x$ "

Then: 1 is related to 2

2 is related to 4

3 is related to 6

Therefore: $R = \{(1,2), (2,4), (3,6)\}$

Notice what happened. $A \times B$ contains all possible ordered pairs. The relation keeps only those pairs that satisfy the chosen rule.

A Useful Analogy

Imagine a classroom.

Every student could potentially sit beside every other student.

That is similar to a Cartesian Product.

Now suppose we only record students who sit in the same row.

We have selected certain connections and ignored others.

That selection is a relation.

NOTES

Domain, Codomain and Range

These three ideas are frequently confused.
Understanding them clearly will help throughout the chapter.

Domain

The Domain consists of all first elements appearing in the relation.

Example: $R = \{(1,4), (2,5), (3,5)\}$

Domain: $\{1,2,3\}$

The Domain answers the question:

"What values are actually participating as inputs?"

Range

The Range consists of all second elements that are actually connected.

For the same relation: $R = \{(1,4), (2,5), (3,5)\}$

Range: $\{4,5\}$

Notice that 5 appears only once in the Range even though it occurs in two ordered pairs. Sets never repeat elements.

Codomain

The Codomain is the entire target set from which outputs may come.

Suppose: $A = \{1,2,3\}$ $B = \{4,5,6,7\}$ and

$R = \{(1,4), (2,5), (3,5)\}$

Then:

Domain = $\{1,2,3\}$

Range = $\{4,5\}$

Codomain = $\{4,5,6,7\}$

The Most Common Confusion

Students often think: Range = Codomain

This is not always true. Range contains only actual outputs.

Codomain contains all possible outputs.

They become equal only in special situations called Onto Functions.

Remember this carefully.

It is one of the most frequently tested conceptual points in board examinations.

NOTES

Inverse Relation

If: $(a,b) \in R$
then: $(b,a) \in R^{-1}$

In simple words:

An inverse relation reverses every connection.

Important Result

Domain of R = Range of R^{-1}

Range of R = Domain of R^{-1}

This result is often useful in objective questions.

Composition of Relations

Mathematics often studies chains of relationships.

Suppose: A is connected to B and B is connected to C

Can we directly connect A to C?

Composition of Relations answers this question.

Example

Suppose: $R = \{(1,2), (2,3)\}$ and $S = \{(2,5), (3,6)\}$

Using the intermediate elements, we obtain:

$$S \circ R = \{(1,5), (2,6)\}$$

The connection passes through the middle set.

Why Composition Matters

Composition allows mathematics to build complex relationships from simpler ones.

Many advanced ideas in calculus, algebra, computer science and machine learning are based on repeated compositions of functions and relations.

What begins as a simple chapter in Class 12 eventually grows into one of the most powerful ideas in modern mathematics.

Reflection

Relations teach us that mathematics is not merely concerned with objects.

It is concerned with connections. A relation tells us which elements are linked.

The Domain tells us where those links begin.

The Range tells us where they end.

The Inverse Relation reverses them.

The Composition of Relations combines them.

Understanding these ideas deeply makes the transition to Functions much easier.

And Functions are where relations become precise.

NOTES

Types of Relations

Not all relations behave in the same way.

Some relations describe self-connections.

Some describe mutual connections.

Some describe chains of connections.

To understand relations more deeply, mathematicians classify them into different types.

Let us explore the most important ones.

Reflexive Relation

A relation R on a set A is called Reflexive if every element of the set is related to itself.

In symbols: $(a,a) \in R$ for every $a \in A$

Example

Let: $A = \{1,2,3\}$ and

$R = \{(1,1), (2,2), (3,3)\}$

Since every element is related to itself, R is Reflexive.

Common Mistake

Students often check only one element.

Reflexivity requires every element of the set to be related to itself.

Symmetric Relation

A relation R is called Symmetric if: Whenever: $(a,b) \in R$ then: $(b,a) \in R$ must also belong to R .

Example

$R = \{(1,2), (2,1), (3,3)\}$

This relation is symmetric because every connection has its reverse connection.

Common Mistake

Students sometimes see one symmetric pair and conclude the entire relation is symmetric.

Every pair must satisfy the condition.

NOTES

Anti-Symmetric Relation

A relation R is called Anti-Symmetric if:

Whenever: $(a,b) \in R$ and $(b,a) \in R$
then: $a = b$

Understanding the Idea

In an anti-symmetric relation,
two different elements cannot point to each other simultaneously.

If both directions exist,
the elements must actually be the same.

Real-Life Analogy

Consider: "less than or equal to"

If: $a \leq b$ and $b \leq a$
then: $a = b$

This is anti-symmetric behaviour.

Example

$R = \{(1,1), (2,2), (1,2)\}$ This relation is anti-symmetric.

Why?

Because there is no pair:
 $(2,1)$

to create a two-way connection between different elements.

Important Note

Anti-Symmetric does NOT mean "not symmetric."

Many students confuse these ideas.

A relation can be:

Neither symmetric nor anti-symmetric,

Only symmetric,

Only anti-symmetric,

Or both.

NOTES

Transitive Relation

A relation R is called Transitive if:
Whenever: $(a,b) \in R$ and $(b,c) \in R$
then: $(a,c) \in R$
must also belong to R .

Understanding the Idea

A transitive relation allows connections to travel through intermediate elements.

If one connection leads to another, the final connection must also exist.

Real-Life Analogy

Suppose: A is taller than B and B is taller than C
Then: A is taller than C The conclusion naturally follows.

Example

$R = \{(1,2), (2,3), (1,3)\}$
Since: $(1,2)$ and $(2,3)$
imply $(1,3)$ the relation is transitive.

Identity Relation

An Identity Relation contains only self-connections.
For a set: $A = \{1,2,3\}$
Identity Relation is: $I = \{(1,1), (2,2), (3,3)\}$

Understanding the Idea

Every element is connected only to itself. Nothing else.

Important Observation

Identity Relations are always:
✓ Reflexive
✓ Symmetric
✓ Transitive

Equivalence Relation

A relation is called an Equivalence Relation if it is:
✓ Reflexive
✓ Symmetric
✓ Transitive
at the same time.

NOTES

Summary Table

Relation Type	Key Idea
Reflexive	Every element relates to itself
Symmetric	Connections work both ways
Anti-Symmetric	Two-way connection only for same element
Transitive	Connections can be chained
Identity	Only self-connections
Equivalence	Reflexive + Symmetric + Transitive

Reflection

These different types of relations reveal an important truth.

Not all connections behave in the same way.

Some relationships are mutual.

Some are directional.

Some create chains.

Some describe identity itself.

By studying these patterns, mathematics learns to classify and understand relationships with precision.

And once relationships become sufficiently structured, they evolve into something even more powerful:

Functions.

NOTES

4. Functions – When Relations Become Precise

Relations allow us to describe connections.

Functions take those connections and make them more disciplined.
A function is one of the most important ideas in mathematics because it describes how one quantity depends upon another.

Whenever changing one thing causes another thing to change,
functions appear.

Temperature changes throughout the day.

Distance changes with time.

Population changes over years.

Interest changes with investment.

All of these can be understood through functions.

The Fundamental Idea

A function is a special type of relation.

Not every relation is a function.

But every function is a relation.

The difference lies in a single rule.

The Golden Rule of Functions

Every input must have exactly one output.

This is the most important sentence in the entire chapter.

Read it again. Every input must have exactly one output.

A Real-Life Example

Consider a school database. Student $\rightarrow$ Roll Number

Every student has exactly one roll number.

Therefore this relationship behaves like a function.

Now consider: Student $\rightarrow$ Favourite Subject

A student may like Mathematics and Physics simultaneously.

One input may produce multiple outputs. Therefore it may not behave
like a function.

Domain, Codomain and Range Revisited

For a function: $f : A \rightarrow B$

A is called the Domain.

B is called the Codomain.

Actual outputs form the Range.

Remember: $\text{Range} \subseteq \text{Codomain}$

NOTES

Types of Functions

One-One Function (Injective Function)

A function is called One-One if different inputs produce different outputs.

In simple words:

No two inputs are allowed to share the same output.

Visual Idea

Imagine assigning unique Aadhaar numbers.

Every person receives a different number.

No duplication is allowed.

That is the spirit of a One-One Function.

Quick Test

If: $a = b$ implies $f(a) = f(b)$ then the function is One-One.

Many-One Function

A function is called Many-One if multiple inputs can produce the same output.

Important Note

Students often believe Many-One Functions are invalid.

They are completely valid functions.

The Golden Rule only says:

One input cannot have multiple outputs.

It does NOT say:

One output cannot have multiple inputs.

Onto Function (Surjective Function)

An Onto Function uses every element of the codomain.

Nothing is left unused.

Key Observation

For an Onto Function:

Range = Codomain

This is one of the most important results in the chapter.

Into Function

An Into Function does not use every element of the codomain.

Some outputs remain unused.

Important Result

For an Into Function:

Range $\subset$ Codomain

The Range is a proper subset of the Codomain.

NOTES

Bijjective Function

A function that is both:
✓ One-One and ✓ Onto
is called a Bijjective Function.

Why Is It Important?

Bijjective Functions are special because they possess inverses.
Whenever a function is bijective, we can reverse the process perfectly.
This idea becomes extremely important in higher mathematics.

Identity Function

An Identity Function maps every element to itself.
 $f(x) = x$ for every x in the Domain.
Every element retains its own identity.

Constant Function

A Constant Function sends every input to the same output.
A Constant Function creates a horizontal line on a graph.
This idea becomes important later in calculus.

Common Confusions

Confusion #1 Every Function is a Relation. TRUE.
 Every Relation is a Function. FALSE.

Confusion #2 One-One and Onto are different properties.
 A function may be:
 ✓ One-One but not Onto
 ✓ Onto but not One-One
 ✓ Both
 ✓ Neither

Confusion #3 Many-One Functions are still Functions.
 Many students incorrectly reject them.
 Remember:
 Many inputs → One output
 Allowed.
 One input → Many outputs
 Not Allowed.

NOTES

Reflection

Functions are among the most powerful ideas ever created in mathematics.

A function tells us how one quantity depends upon another. It transforms mathematics from a collection of objects into a study of change, dependence and structure.

Every graph, every equation, every model and every prediction ultimately rests upon this simple idea:

One input.

Exactly one output.

That is the heart of a function.

5. Composition and Inverse Functions

In mathematics, functions rarely exist in isolation.

Often, the output of one function becomes the input of another.

When this happens, we combine functions through a process called composition.

Similarly, if a function can be reversed, we obtain an inverse function. Both ideas are central to advanced mathematics and appear repeatedly in calculus, algebra and computer science.

Composition of Functions

Suppose: $f: A \rightarrow B$ and $g: B \rightarrow C$

The output of f becomes the input of g .

We can therefore combine them into a single function.

This new function is called the composition of functions.

It is written as: $(g \circ f)(x)$

and read as: "g composed with f"

Important Observation

Function composition is generally not commutative.

That means: $(g \circ f)(x) \neq (f \circ g)(x)$ in most cases.

Real-Life Analogy

Imagine:

Student $\rightarrow$ Roll Number

Roll Number $\rightarrow$ Examination Result

Combining both relationships gives:

Student $\rightarrow$ Examination Result

This is the essence of composition.

NOTES

Inverse Functions

An inverse function reverses the action of a function.

If a function takes us from A to B,
its inverse takes us from B back to A.

Condition for Existence

Not every function has an inverse.

A function possesses an inverse only when it is:

Bijjective That means:

✓ One-One

✓ Onto

Why?

Because every output must correspond to exactly one input.

Otherwise the reverse process becomes ambiguous.

Real-Life Analogy

Think of locking and unlocking a door.

Locking: Function

Unlocking: Inverse Function

One process reverses the other.

6. Binary Operations

Binary Operations form the final topic of this chapter.

The idea is surprisingly simple.

A Binary Operation combines two elements and produces a third element.

Familiar Examples

Addition $3 + 4 = 7$

Multiplication $2 \times 5 = 10$

Subtraction $8 - 3 = 5$

These are all binary operations.

Why?

Because two inputs are combined to produce one output.

Formal Definition

A binary operation on a set A is a function:

$$A \times A \rightarrow A$$

This means:

Two elements from A are combined,
and the result must also belong to A.

NOTES

Example

Consider: $A = \{0,1,2\}$
Suppose an operation is defined on A .
For the operation to be binary,
every result must remain inside A .
If any result falls outside the set,
the operation is not a binary operation on that set.

Closure Property

This idea is called Closure.

A set is closed under an operation if every result remains inside the set.

Example

Integers are closed under addition.

$$3 + 4 = 7$$

All are integers.

Therefore addition is a binary operation on integers.

Counter Example

Natural numbers are not closed under subtraction.

$$3 - 5 = -2$$

But -2 is not a natural number.

Therefore subtraction is not a binary operation on natural numbers.

Why Binary Operations Matter

Binary Operations become the foundation of:

- ✓ Algebra
- ✓ Abstract Mathematics
- ✓ Group Theory
- ✓ Linear Algebra
- ✓ Modern Cryptography

Although the topic appears small in Class 12, its influence extends far beyond school mathematics.

NOTES

Most Common Examination Mistakes

Mistake 1

Confusing Range with Codomain.

Remember:

$\text{Range} \subseteq \text{Codomain}$

Always.

Mistake 2

Believing every relation is a function.

False.

Only special relations become functions.

Mistake 3

Forgetting that composition depends on order.

Usually: $g \circ f \neq f \circ g$

Mistake 4

Assuming every function has an inverse.

Only bijective functions possess inverses.

Mistake 5

Rejecting Many-One functions.

Many-One functions are perfectly valid functions.

NOTES

Final Reflection

At first glance, Relations and Functions appear to be a chapter about definitions and symbols.

But beneath those definitions lies a deeper idea.

Mathematics is a study of relationships.

Numbers matter because they relate to one another.

Variables matter because they depend upon one another.

Functions matter because they describe how change occurs.

This chapter teaches us that understanding begins not with isolated objects, but with connections.

And perhaps that is true not only in mathematics, but in life itself.